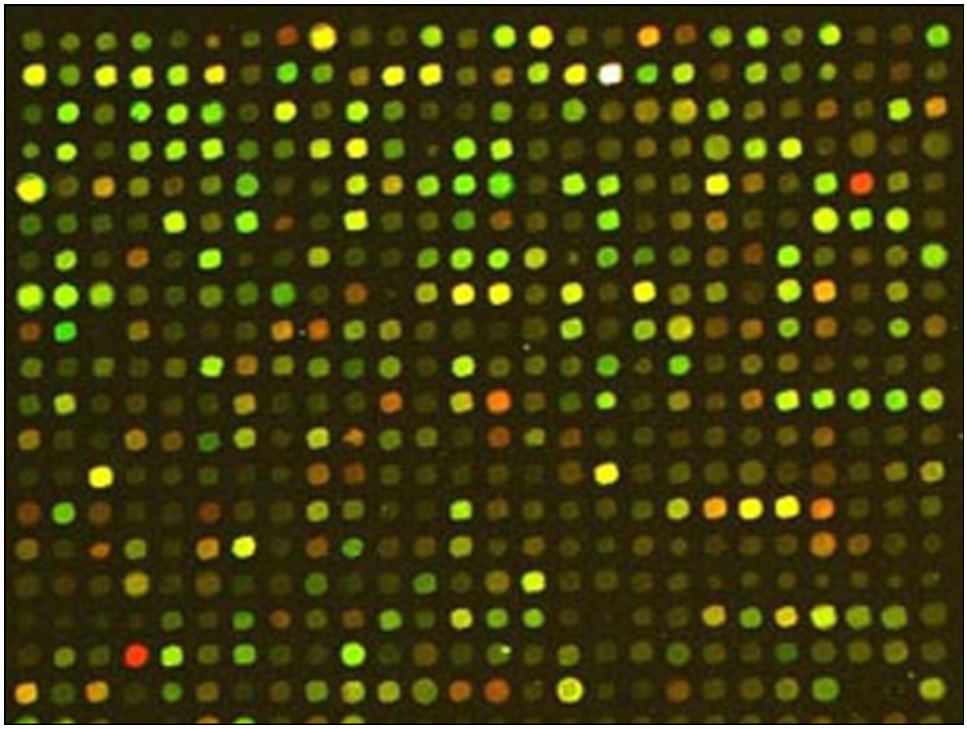
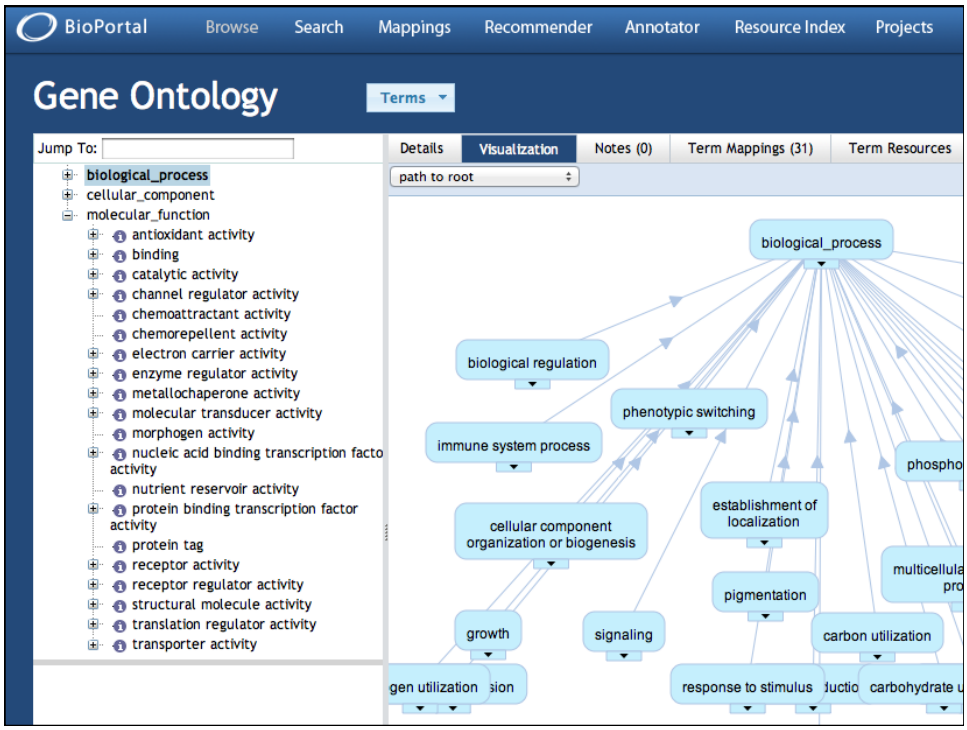
NATIONAL CENTER FOR
BIOMEDICAL ONTOLOGY

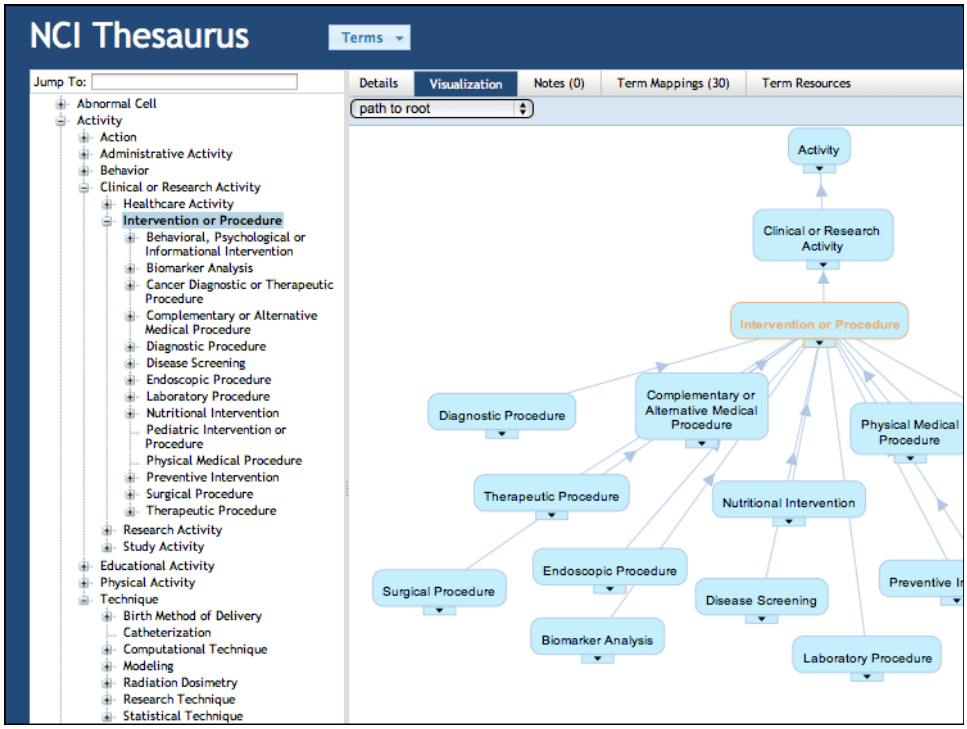
Mark Musen and the NCBO team
Stanford University

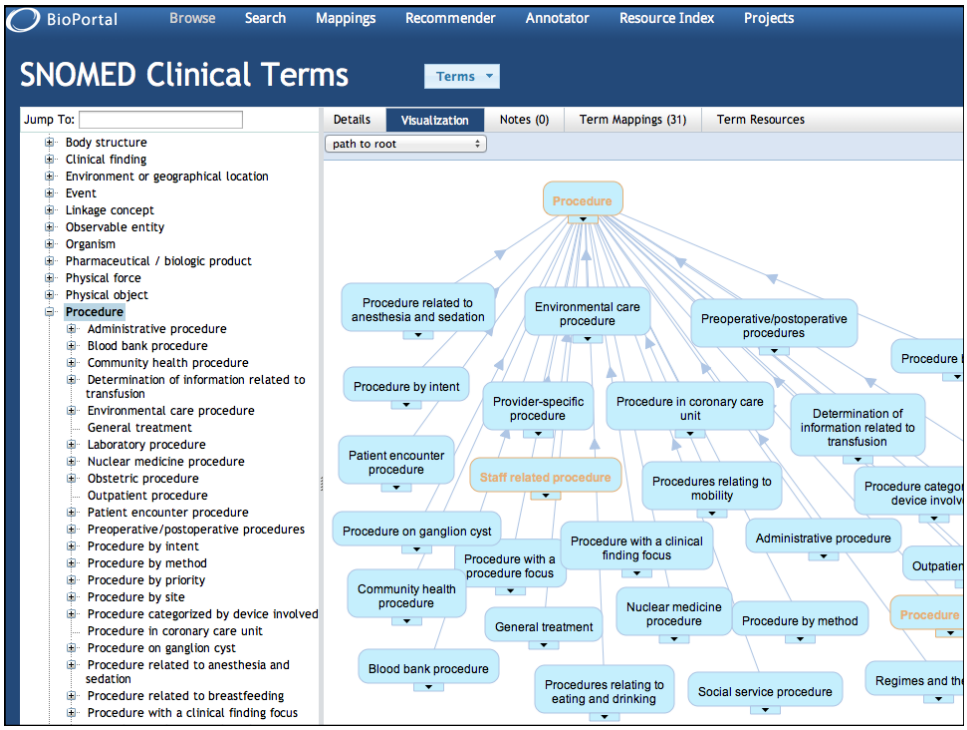
Semantic Technology Goes Mainstream: The NCBO Experience






BioPortal
Browse
Search
Mappings
Recommender
Annotator
Resource Index
Projects
Gene Ontology
Terms
Jump To:
biological_process
cellular_component
molecular_function
antioxidant activity
binding
catalytic activity
channel regulator activity
chemoattractant activity
chemorepellent activity
electron carrier activity
enzyme regulator activity
metallochaperone activity
molecular transducer activity
morphogen activity
nucleic acid binding transcription factor activity
nutrient reservoir activity
protein binding transcription factor activity
protein tag
receptor activity
receptor regulator activity
structural molecule activity
translation regulator activity
transporter activity
Details
Visualization
Notes (0)
Term Mappings (31)
Term Resources
path to root
biological_process
biological regulation
phenotypic switching
immune system process
cellular component organization or biogenesis
establishment of localization
pigmentation
growth
signaling
carbon utilization
response to stimulus

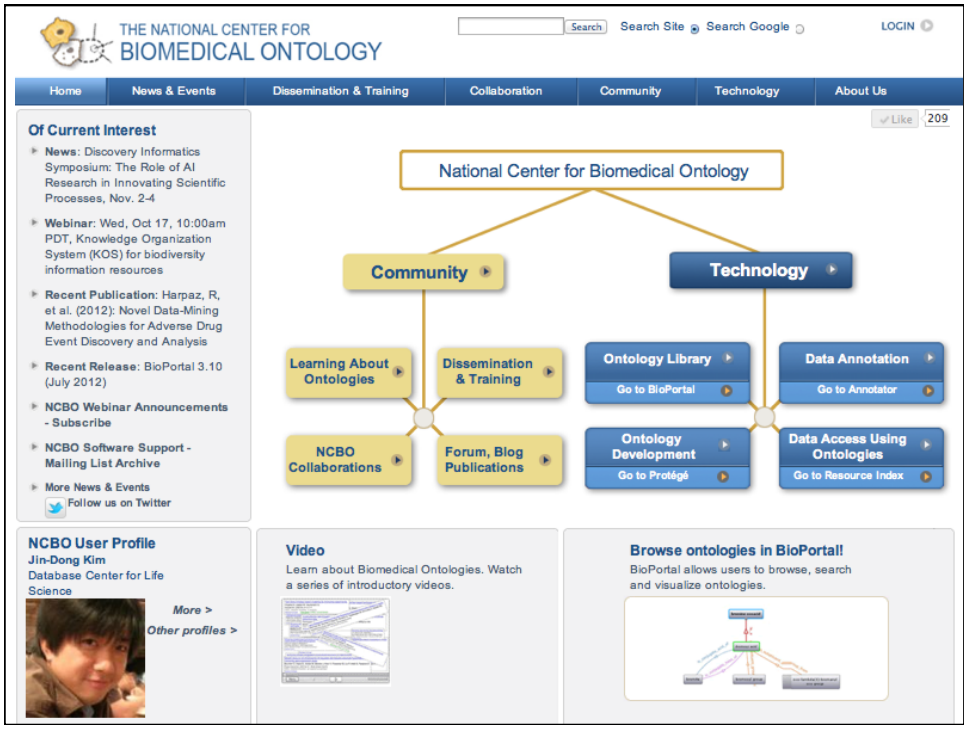

NCI Thesaurus
Terms
Jump To:
Abnormal Cell
Activity
Action
Administrative Activity
Behavior
Clinical or Research Activity
Healthcare Activity
Intervention or Procedure
Behavioral, Psychological or Informational Intervention
Biomarker Analysis
Cancer Diagnostic or Therapeutic Procedure
Complementary or Alternative Medical Procedure
Diagnostic Procedure
Disease Screening
Endoscopic Procedure
Laboratory Procedure
Nutritional Intervention
Pediatric Intervention or Procedure
Physical Medical Procedure
Preventive Intervention
Surgical Procedure
Therapeutic Procedure
Research Activity
Study Activity
Educational Activity
Physical Activity
Technique
Birth Method of Delivery
Catheterization
Computational Technique
Modeling
Radiation Dosimetry
Research Technique
Statistical Technique
Details
Visualization
Notes (0)
Term Mappings (30)
Term Resources
path to root
Activity
Clinical or Research Activity
Intervention or Procedure
Diagnostic Procedure
Complementary or Alternative Medical Procedure
Physical Medical Procedure
Therapeutic Procedure
Nutritional Intervention
Endoscopic Procedure
Surgical Procedure
Disease Screening
Biomarker Analysis
Laboratory Procedure


BioPortal
Browse
Search
Mappings
Recommender
Annotator
Resource Index
Projects
SNOMED Clinical Terms
Terms
Jump To:
Body structure
Clinical finding
Environment or geographical location
Event
Linkage concept
Observable entity
Organism
Pharmaceutical / biologic product
Physical force
Physical object
Procedure
Administrative procedure
Blood bank procedure
Community health procedure
Determination of information related to transfusion
Environmental care procedure
General treatment
Laboratory procedure
Nuclear medicine procedure
Obstetric procedure
Outpatient procedure
Patient encounter procedure
Preoperative/postoperative procedures
Procedure by intent
Procedure by method
Procedure by priority
Procedure by site
Procedure categorized by device involved
Procedure in coronary care unit
Procedure on ganglion cyst
Procedure related to anesthesia and sedation
Procedure related to breastfeeding
Procedure with a clinical finding focus
Details
Visualization
Notes (0)
Term Mappings (31)
Term Resources
path to root
Procedure
Procedure related to anesthesia and sedation
Environmental care procedure
Preoperative/postoperative procedures
Procedure by intent
Provider-specific procedure
Procedure in coronary care unit
Determination of information related to transfusion
Patient encounter procedure
Staff related procedure
Procedures relating to mobility
Procedure on ganglion cyst
Procedure with a clinical finding focus
Administrative procedure
Procedure with a procedure focus
Community health procedure
Nuclear medicine procedure
Procedure by method
General treatment
Blood bank procedure
Procedures relating to eating and drinking
Social service procedure

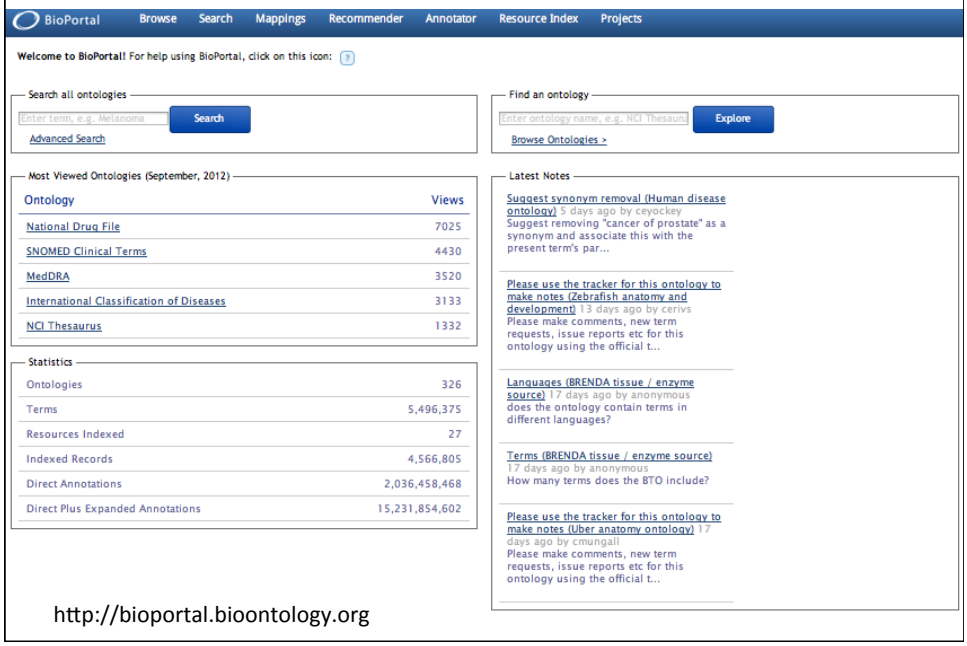
The National Center for Biomedical Ontology

- We **create and maintain a library** of biomedical ontologies and terminologies.
- We **build tools and Web services** to enable the use of ontologies and terminologies.
- We **collaborate with scientific communities** that develop and use ontologies and terminologies in biomedicine.

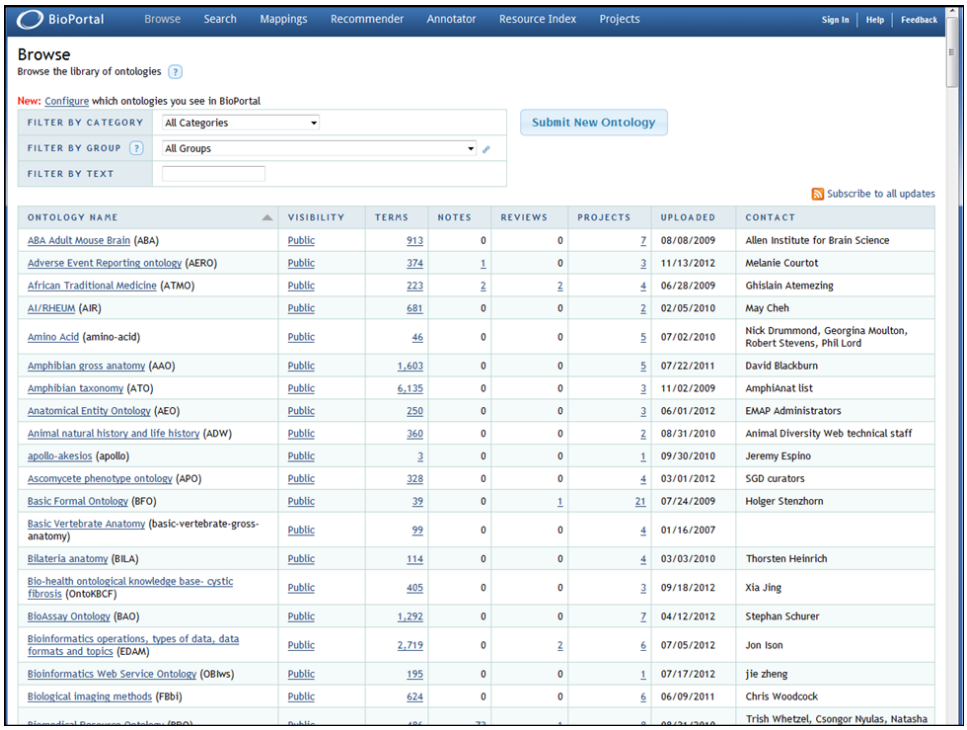
NATIONAL CENTER FOR
BIOMEDICAL ONTOLOGY



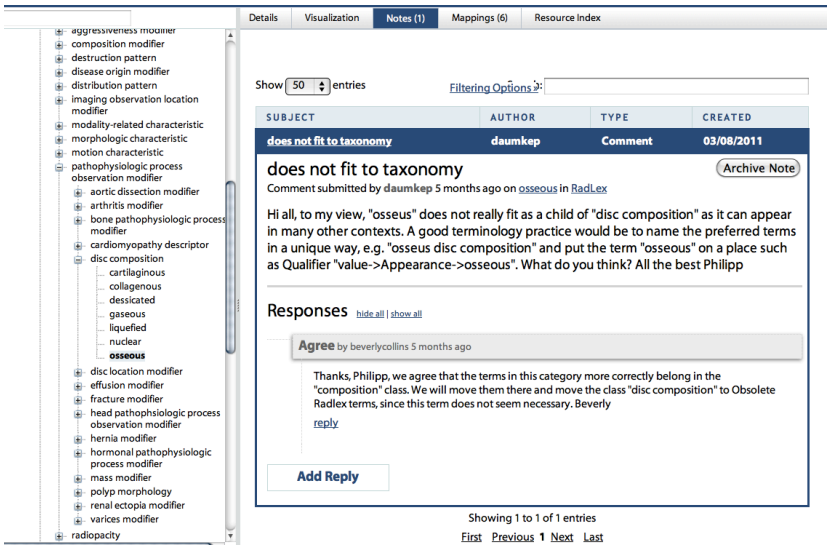
BioPortal Ontology Repository



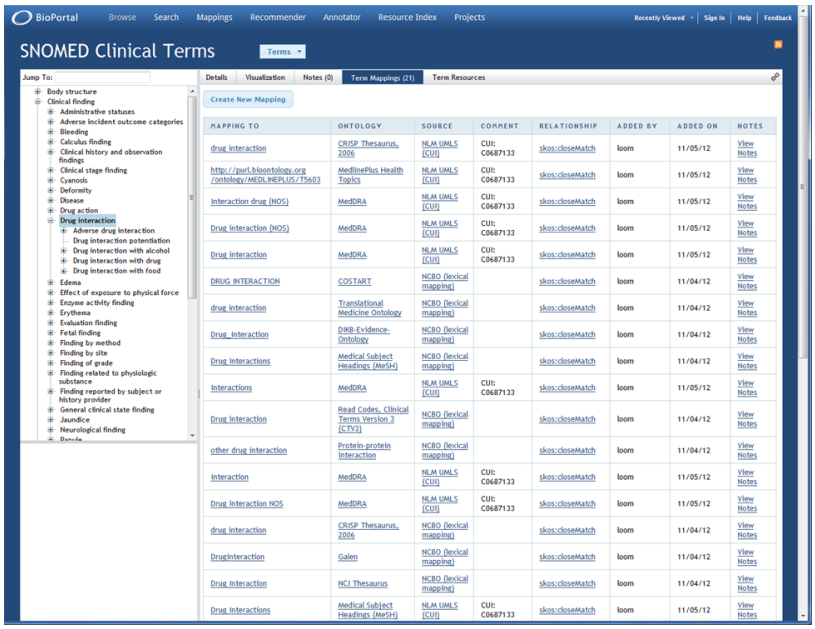
http://bioportal.bioontology.org



Community Comments in BioPortal

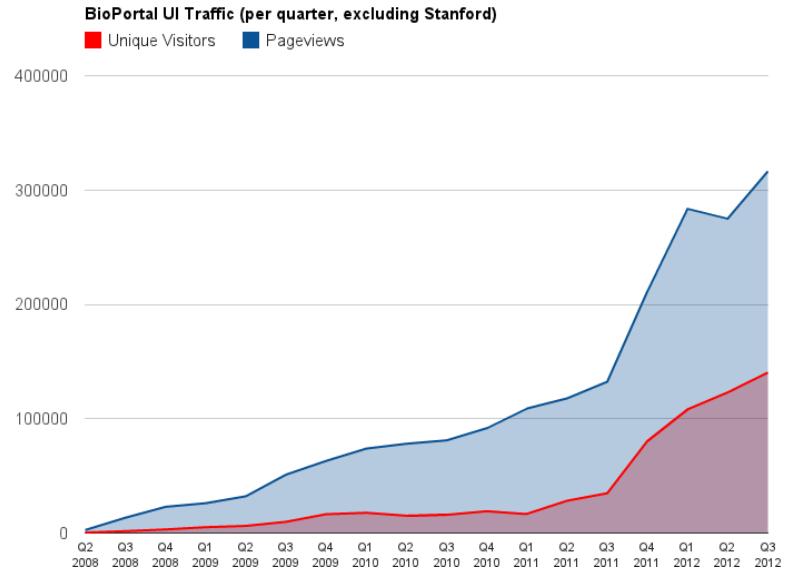


Users can view and create mappings

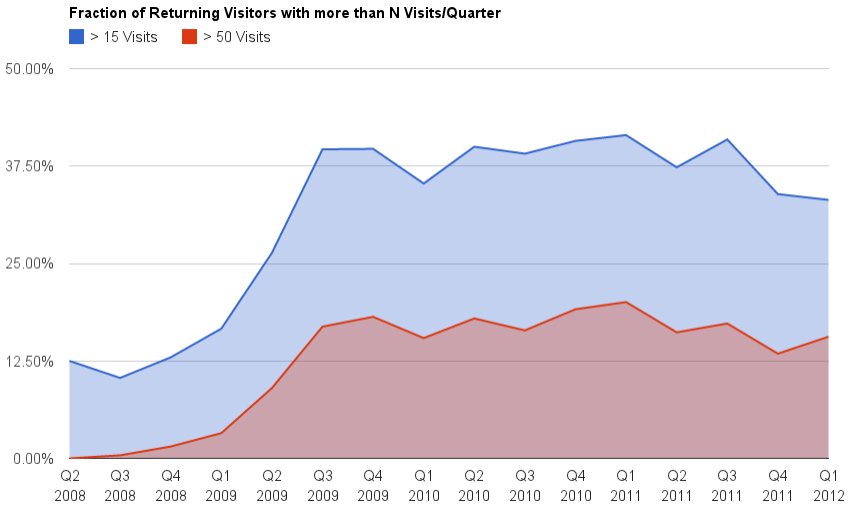


12

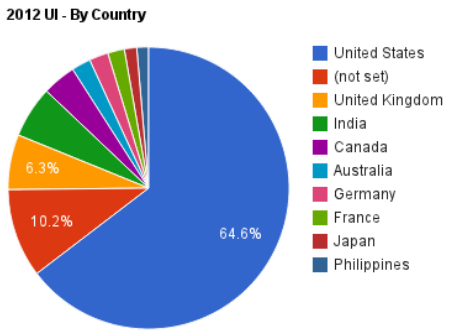
BioPortal Traffic – Volume



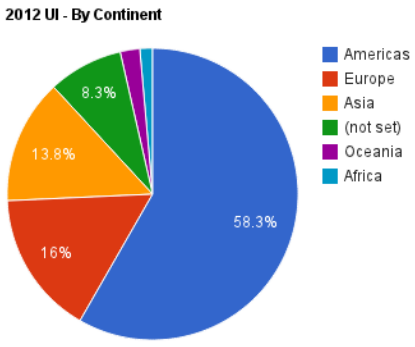
BioPortal Engagement



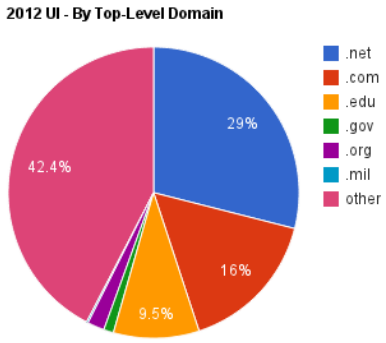
BioPortal Traffic – By Country



BioPortal Traffic – By Continent



BioPortal Traffic – By Internet Domain



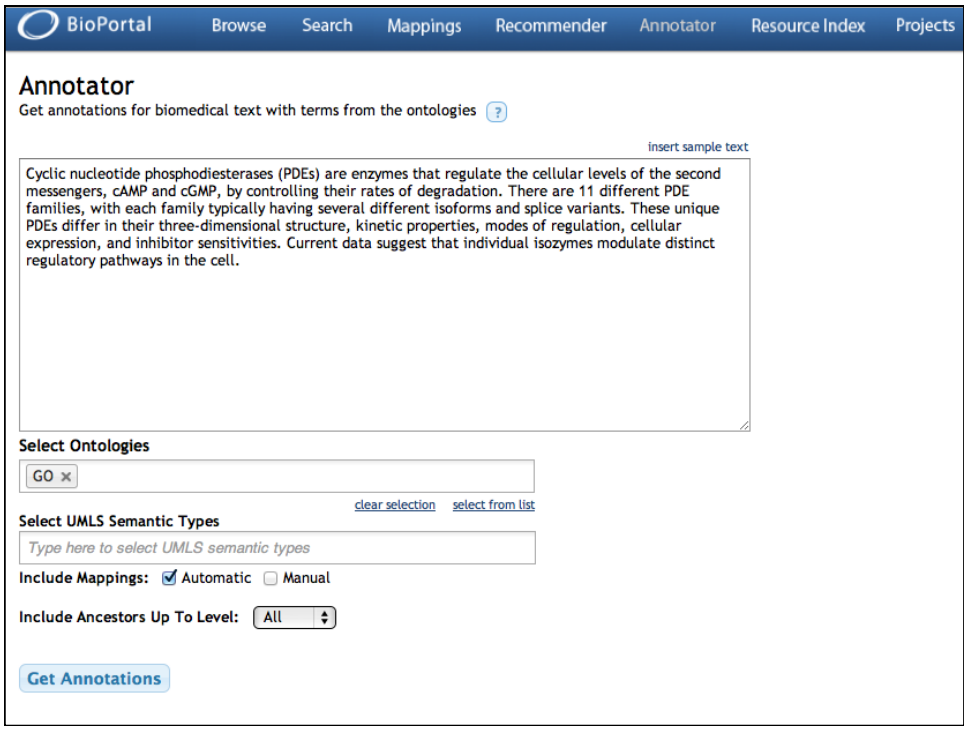
BioPortal UI Traffic – Top Users

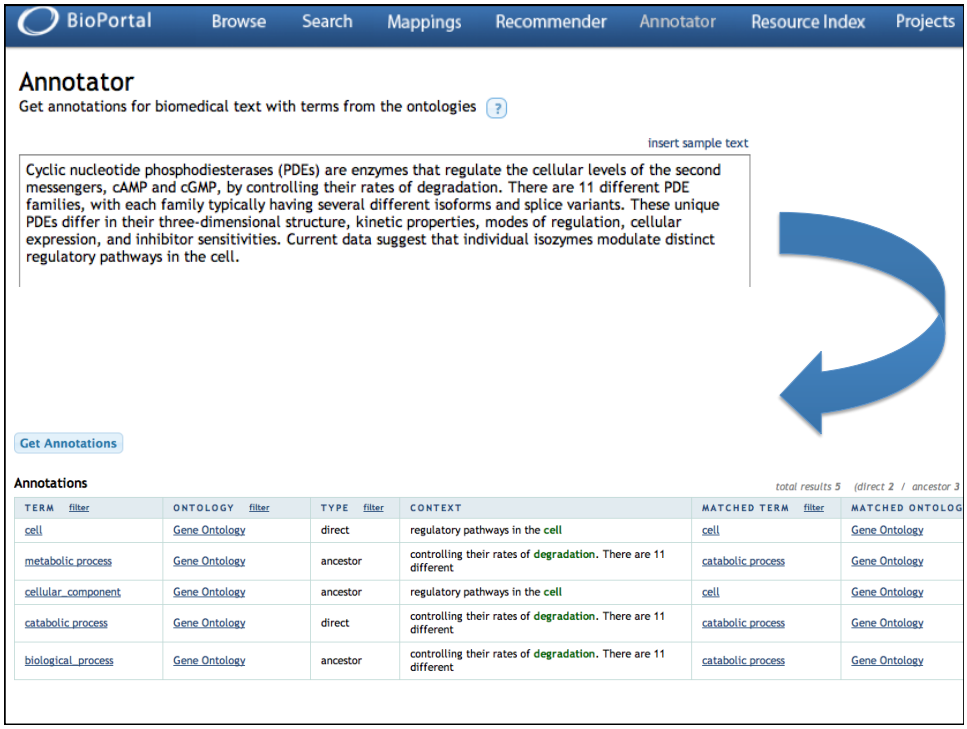
Thomson Financial
Japan Network Information Center
Department of Veterans Affairs
National Health Service (UK)
University of British Columbia
University of Florida
MRC Human Genome Mapping Project Resource Centre
University of California, San Diego
Mayo Foundation for Medical Education and Research
University of California, Davis
University of Pennsylvania

BioPortal is building an online community of users who

- Develop, upload, and apply ontologies
- Map ontologies to one another
- Comment on ontologies via "notes" to give feedback
  - To the ontology developers
  - To one another
- Make proposals for specific changes to ontologies
- Stay informed about ontology changes and proposed changes via "push" technology
- Incorporate BioPortal services into their own technologies

19





NCBO Annotator

- Typically called as Web service by client programs
- Takes as input some text; generates as output links between the text and terms in any or all NCBO ontologies
- Enables high-throughput
  - Annotation of textual metadata
  - Text mining of electronic medical records
  - Concept-based indexing and retrieval of online data sets

BioPortal Browse Search Mappings Recommender Annotator Resource Index Projects

Search Resource Index

Search biomedical resources

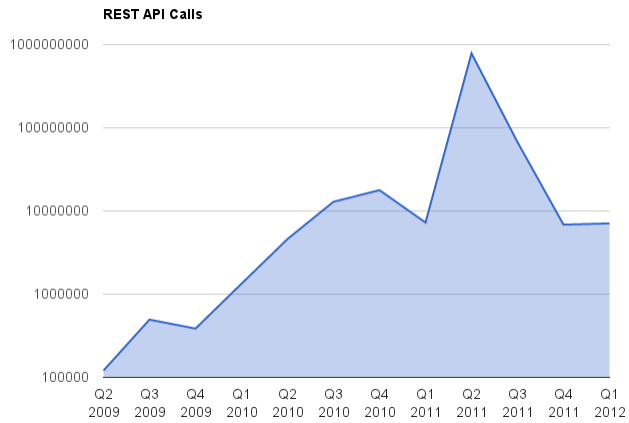
Start typing to find terms to search the index with | Search Resource Index

select ontologies to search

Available Resources

RESOURCE	DESCRIPTION
AgingGenesDB (via NIF)	This database provides a searchable and browsable list of aging related genes and their effects ... [more]
ArrayExpress	ArrayExpress is a public repository for microarray data, which is aimed at storing MIAME-compli ... [more]
ARRS GoldMiner	ARRS GoldMiner provides instant access to images published in selected peer-reviewed radiology ... [more]
Biositemaps	Biositemaps represent a mechanism for computational biologists and bio-informaticians to openly ... [more]
caNanoLab	caNanoLab is a data sharing portal designed to facilitate information sharing in the biomedical ... [more]
Cell Centered Database (via NIF)	The Cell Centered Database is a publicly accessible resource for high resolution 2D, 3D and 4D ... [more]
ClinicalTrials.gov	ClinicalTrials.gov provides regularly updated information about federally and privately support ... [more]
Conserved Domain Database (CDD)	The Conserved Domain Database (CDD) contains protein domain models imported from outside source ... [more]
Database of Genotypes and Phenotypes	The database of Genotypes and Phenotypes (dbGaP) was developed to archive and distribute the re ... [more]
DrugBank	DrugBank is offered to the public as a freely available resource. Use and re-distribution of th ... [more]
Gene Expression Omnibus DataSets	A gene expression/molecular abundance repository supporting MIAME compliant data submissions, ... [more]
MICAD	Molecular Imaging and Contrast Agent Database
ModelDB (via NIF)	ModelDB provides an accessible location for storing and efficiently retrieving computational ne ... [more]
Online Mendelian Inheritance in Man	OMIM is a comprehensive, authoritative, and timely compendium of human genes and genetic phenot ... [more]
Pathway Commons	Pathway Commons is a convenient point of access to biological pathway information collected fro ... [more]

REST Traffic – Volume



REST Traffic – Top Users (2012)

Institution
Buck Institute
NextBio
Software Research & Development and Consultancy, Turkey
Protégé
RSNA
Mayo
Scripps Institute
Universidad de Murcia, ES
EBI, UK
University of Leicester, UK
Emory
University of Amsterdam, NL
INRIA, FR
Oxford e-Research Center, UK
Pacific Northwest National Labs
ISI
UCSF
Partners



Stanford Ontology Tools

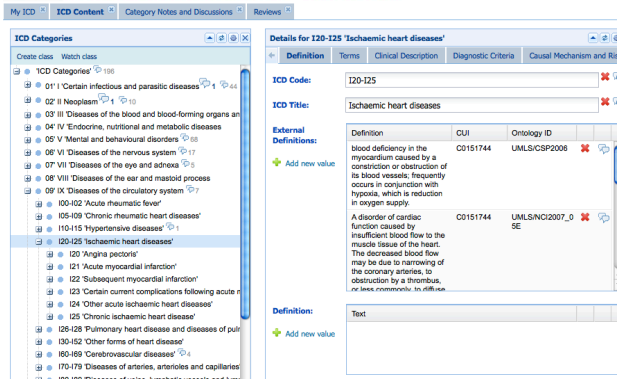
1. BioPortal: A community-based ontology repository
2. WebProtégé: A Web-based collaborative ontology editor

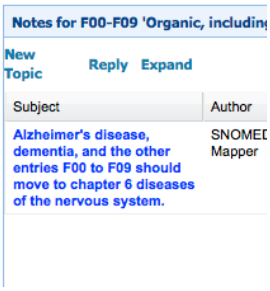
WebProtégé Functionalities

Manage class hierarchy

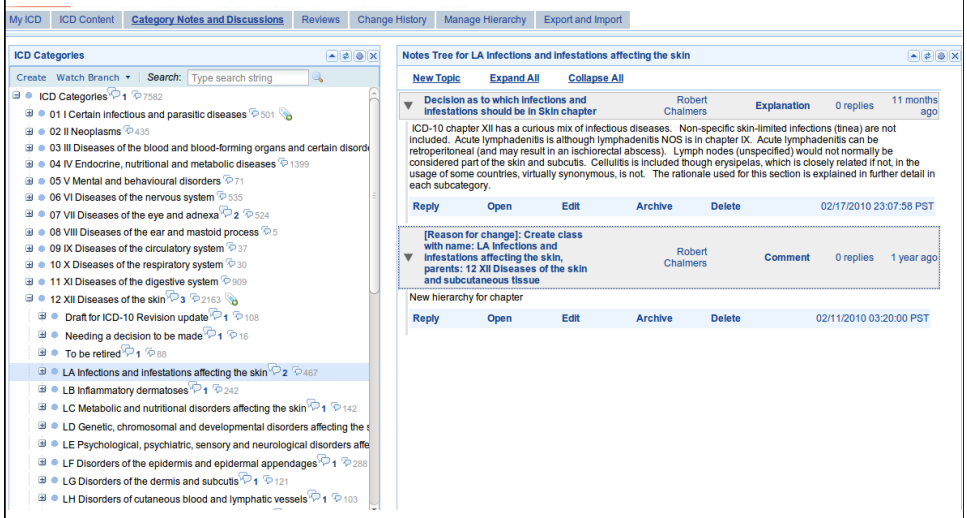
Enter and edit properties

Add notes and discussions

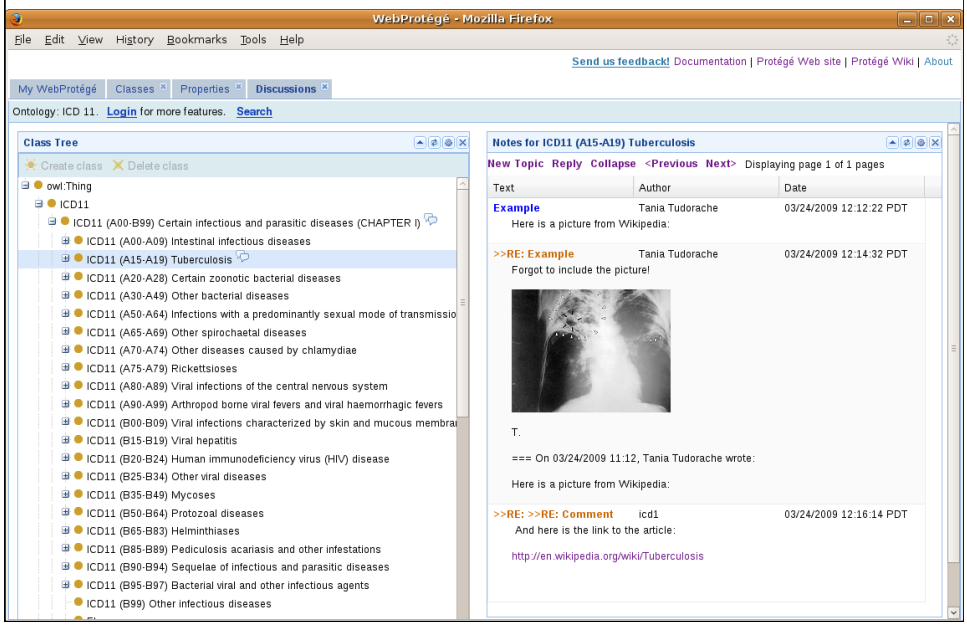




Like BioPortal, WebProtégé supports notes and threaded discussions



As with BioPortal, notes may include multimedia



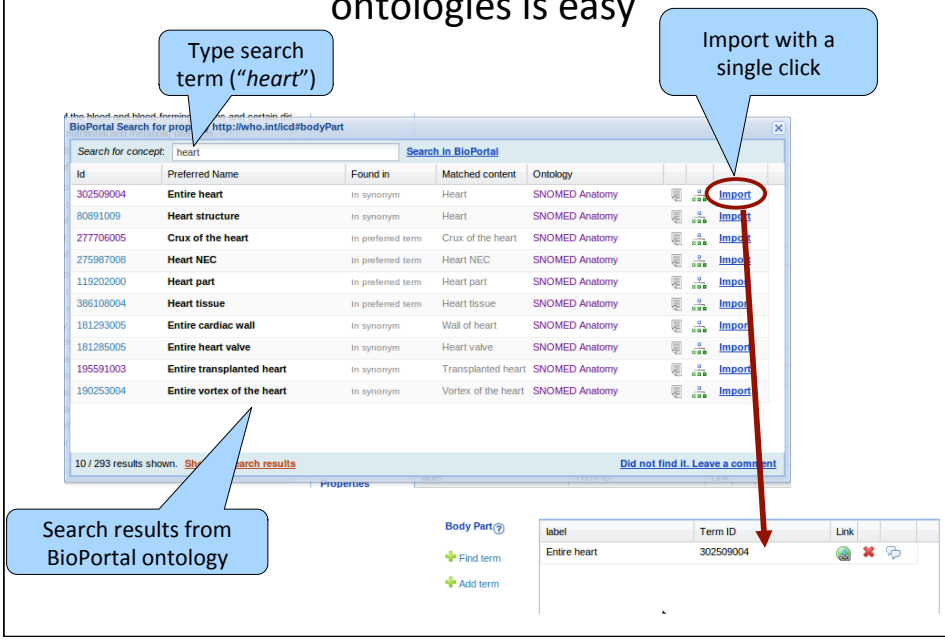
Integration of Ontology Authoring, Publishing, and Peer Review

Our Goal: Enable users to go seamlessly between ontology authoring, editing, publishing, and peer review

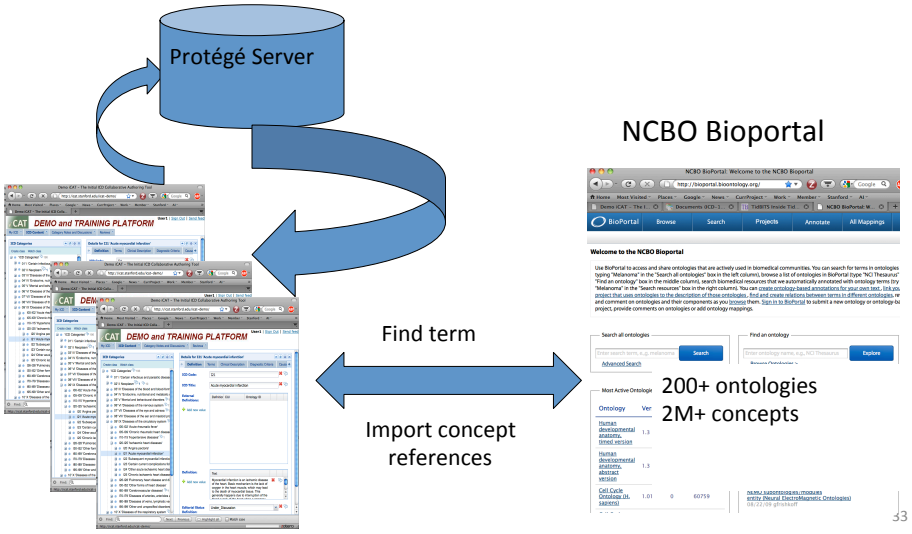
- Notes created in BioPortal are visible in WebProtégé
- User accounts and roles are shared between BioPortal and WebProtégé
- Users don't need to be aware of the difference: they just get their work done



Importing terms from BioPortal ontologies is easy



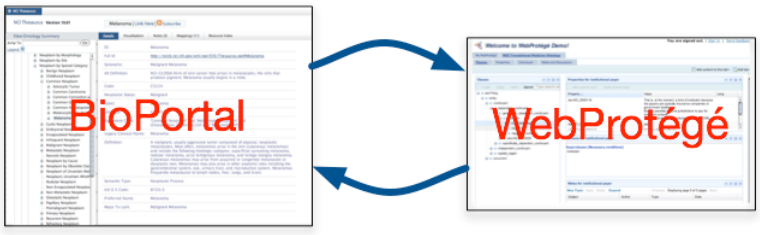
Authoring new ontologies benefits from existing ontologies stored in Bioportal



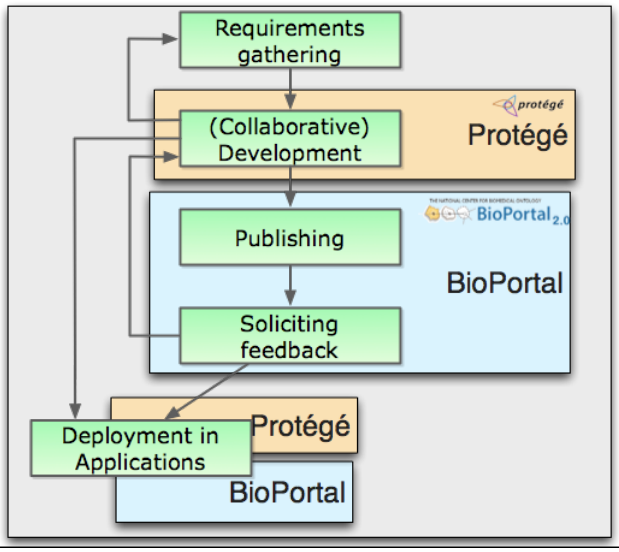
Integration of Ontology Authoring, Publishing, and Peer Review

Our Goal: Enable users to go seamlessly between ontology authoring, editing, publishing, and peer review

- Notes created in BioPortal are visible in WebProtégé
- User accounts and roles are shared between BioPortal and WebProtégé
- Users don't need to be aware of the difference: they just get their work done



A Simplified Ontology Lifecycle



Our initial proposal for NCBO did not predict …

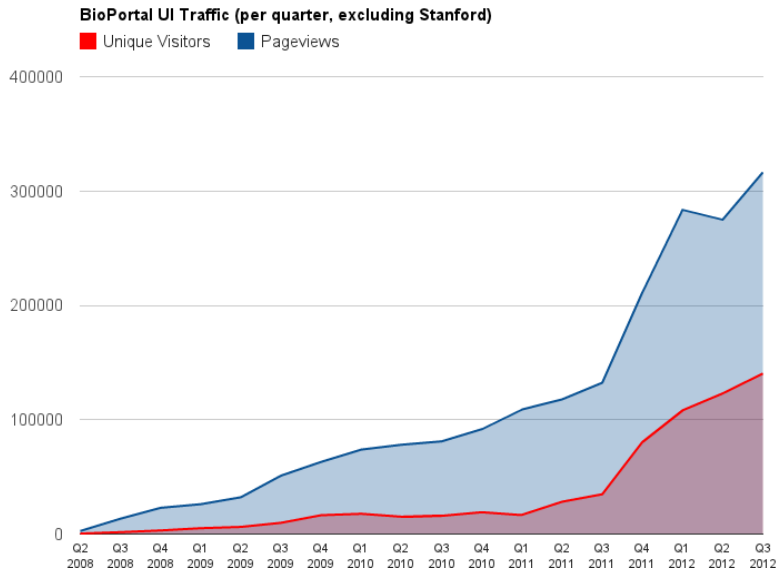
- The exponential increase in the use of semantic technology in biomedicine
- The huge user community that NCBO would attract
- The time that it would take for our technology to become sufficiently robust for other investigators to stake their success on us
- All the applications that investigators would find for our technology
- Opportunities for integrating ontology authoring with dissemination
- That people would take us for granted

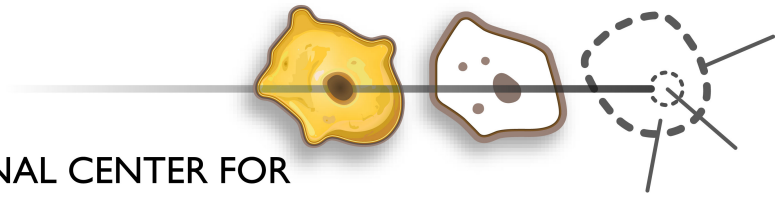
The National Center for Biomedical Ontology

- We **create and maintain a library** of biomedical ontologies and terminologies.
- We **build tools and Web services** to enable the use of ontologies and terminologies.
- We **collaborate with scientific communities** that develop and use ontologies and terminologies in biomedicine.

NATIONAL CENTER FOR
BIOMEDICAL ONTOLOGY

BioPortal Traffic – Volume





NATIONAL CENTER FOR

BIOMEDICAL ONTOLOGY

http://bioontology.org